

Oren Sultan, AI Research Scientist, Computer Science PhD Candidate

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1 Education

- **Feb 2023 - Feb 2027:** PhD Candidate (**direct track for excellent students**) in Computer Science at The Hebrew University, Israel. Advisor: Prof. Dafna Shahaf. **In progress.**
- **Oct 2020 - Dec 2022:** M.Sc in Computer Science, The Hebrew University, Israel. Advisor: Prof. Dafna Shahaf. **GPA: 96.5/100.** Two long papers were accepted to the main conference of **EMNLP 2022**. **Selected to deliver graduation speech.**
 - **May 2022 - Sep 2022: Erasmus+ Exchange Honors Program & Scholarship**, The Technical University of Munich.
- **Oct 2015 - Mar 2019:** B.Sc in Computer Science, The Hebrew University, Israel. **GPA: 90.21/100, Magna Cum Laude.**
 - **Feb 2018 - Jun 2018:** Student Exchange **Honors Program & Scholarship**, The University of Melbourne.

2 Honors & Awards, Scholarships & Fellowships

- **2024 - present: Fellowship for outstanding PhD candidates in AI and Data Science**, The Planning & Budgeting Committee (PBC) of Council For Higher Education (CHE), Israel.
- **2022 - present: MSc & PhD Scholarship** – Dafna Shahaf Hyadata-Lab, The Hebrew University, Israel.
- **2022: Erasmus+ Scholarship for honors students**, Technical University of Munich, Germany.
- **2019:** B.Sc in Computer Science – **Magna Cum Laude**, The Hebrew University of Jerusalem, Israel.
- **2018: Scholarship for honors students**, The University of Melbourne, Australia.
- **2013: Annual Excellence Tech Project Award**, The Israeli Intelligence Corps, Commander of Unit 8200 (rank: Tat Aluf).
- **2012: Annual Excellence Team Award**, The Israeli Intelligence Corps, Commander of R&D, Unit 8200 (rank: Aluf Mishne).

3 Peer-Reviewed Publications

- **2026:** HaltEval: Evaluating Non-Termination Reasoning in AI Coding Agents on Real-World Repositories
 - **Oren Sultan**, Julien Vanegue, Yossi Adi, Peter O’Hearn. Submitted to AAAI 2027.
- **2026:** LLMs versus the Halting Problem: Revisiting Program Termination Prediction
 - **Oren Sultan**, Jordi Armengol-Estape, Pascal Kesseli, Julien Vanegue, Dafna Shahaf, Yossi Adi, Peter O’Hearn. **EMNLP 2026 Findings.**
- **2025:** CWM: An Open-Weights LLM for Research on Code Generation with World Models
 - Meta FAIR CodeGen, **Oren Sultan**. Technical report.
- **2025:** A Neuro-Symbolic Approach for Reliable Proof Generation with LLMs
 - **Oren Sultan**, Eitan Stern, Dafna Shahaf. **EMNLP 2026 Findings**, ISCOL 2025.
- **2024:** Visual Editing with LLM-based Tool Chaining: An Efficient Distillation Approach for Real-Time Applications
 - **Oren Sultan**, Alex Khasin, Guy Shiran, Asnat Greenstein-Messica, Dafna Shahaf. **EMNLP 2024.**
- **2024:** ParallelPARC: A Scalable Pipeline for Generating Natural-Language Analogies
 - **Oren Sultan**, Yonatan Bitton, Ron Yosef, Dafna Shahaf. **NAACL 2024**, ISCOL 2024.

- **2022:** Life is a Circus and We are the Clowns: Automatically Finding Analogies between Situations and Processes
– **Oren Sultan**, Dafna Shahaf. **EMNLP 2022**.
- **2022:** Breakpoint Transformers for Modeling and Tracking Intermediate Beliefs
– Kyle Richardson, Ronen Tamari, **Oren Sultan**, Reut Tsarfaty, Dafna Shahaf, Ashish Sabharwal. **EMNLP 2022**.

4 Professional Experience

- **07.26 - present:** Research Scientist Intern, Google Research.
- **03.26 - 06.26:** Research Scientist (part-time), Meta Superintelligence Labs, FAIR.
- **07.25 - 01.26:** Research Scientist Intern, Meta Superintelligence Labs, FAIR.
- **03.24 - 06.25:** Research Scientist (part-time), Lightricks Research.
- **10.23 - 12.23:** Research Team Leader, Unit 8200 (Captain in Res.).
- **08.23 - 02.24:** Research Scientist Intern, Lightricks Research.
- **03.23 - 04.23:** Visiting Researcher, University of Surrey & Sheffield.
- **.05.22 - 07.22:** NLP Researcher, The Technical University of Munich – Social Computing Lab.
- **11.21 - Present:** MSc & PhD AI / NLP Researcher, The Hebrew University, Israel. **Advisor:** Prof. Dafna Shahaf.
- **08.19 - 09.20:** Backend Software Engineer (Search), Algorithms group, Fiverr.
- **11.16 - 12.17:** Data Engineer Consultant, Madeira Data Solutions.
- **06.14 - 10.15:** Data Engineer, MyThings.
- **2008 - 2013:** MSSQL Data Engineer, Data Engineering Founder & Team Leader – R&D, Unit 8200.

5 Invited Talks

- **Apr 2025:** “Visual Editing with LLM-based Tool Chaining”, Nvidia & NLP IL Meetup, Israel.
- **Sep 2024:** “Visual Editing with LLM-based Tool Chaining”, AI course, Lightricks, Israel.
- **Sep 2024:** “Fine-Tuning Large Language Models (LLMs)”, AI course, Lightricks, Israel.
- **Jun 2023:** “Finding Analogies between Situations and Processes”, MSc graduation ceremony, Hebrew University, Israel.
- **Mar 2023:** “Analogies of Situations and Processes: Detection, Explanation, Generation”, SheffieldNLP, UK.
- **Jan 2023:** “Using LLMs for Natural-Language Analogies Generation”, BIU-NLP, Bar Ilan University, Israel.
- **Jan 2023:** “Automatically Finding Analogies between Situations and Processes”, The Hebrew University, Israel.
- **Jan 2022:** “Breakpoint Transformers for Modeling and Tracking Intermediate Beliefs“, Aristo team, AI2, Seattle, USA.

6 Academic Activities

- **TA:** “self-paced introduction to data science” (67876) at the Center for Interdisciplinary Data-Science Research (CIDR), The Hebrew University, Israel. 2023 - present.
- **Conference Reviewer:** ACL 24, ACL 25, ACL 26, EMNLP 26, AACL 2027

7 Miscellaneous

- **Citizenships:** Israeli, Portuguese.
- **Languages:** Hebrew (mother tongue), English (fluent), Spanish (intermediate), German (elementary).